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2.1.1. List operations

08:32

Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options:

1. Add
2. Remove
3. Display
4. Quit

The program should repeatedly prompt the user to enter a choice from the menu. Depending on the choice selected, the program should perform the following actions:

- **Add:** Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input".
- **Remove:** Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it; otherwise, display "Element not found". If the list is empty, display "List is empty".
- **Display:** Displays the current list of integers. If the list is empty, display "List is empty".
- **Quit:** Exits the program.
- The program should handle invalid menu choices by displaying "Invalid choice". Ensure that the program continues to prompt the user until they choose to quit (option 4).

```
# listOps.py
```

```
int_list = []
```

```
while True:
```

```
    print("1. Add")
```

```
    print("2. Remove")
```

```
    print("3. Display")
```

```
    print("4. Quit")
```

```
    choice = input("Enter choice: ")
```

```
    if choice == '1': # Add
```

```
        value = input("Integer: ")
```

```
        try:
```

```
            num = int(value)
```

```
            int_list.append(num)
```

```
            print("List after adding:", int_list)
```

```
        except ValueError:
```

```
            print("Invalid input")
```

```
    elif choice == '2': # Remove
```

```
        if not int_list:
```

```
            print("List is empty")
```

```
        else:
```

```
            value = input("Integer: ")
```

```
            try:
```

```
    num = int(value)
    if num in int_list:
        int_list.remove(num)
        print("List after removing:", int_list)
    else:
        print("Element not found")
except ValueError:
    print("Invalid input")

elif choice == '3': # Display
    if not int_list:
        print("List is empty")
    else:
        print(int_list)

elif choice == '4': # Quit
    break

else:
    print("Invalid choice")
```

Average time
0.070 s
66.07 ms

Maximum time
0.100 s
100.00 ms

2 out of 2 shown test case(s) passed
1 out of 1 hidden test case(s) passed

Test case 1
Passed
Debug

Expected output

Actual output

1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 1
Integer: 11
List after adding: [11]
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 1
Integer: 20
List after adding: [11, 20]
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 2
Integer: 26
Element not found:
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 2
Integer: 15
List after removing: [26]
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 2
Integer: 20
List after removing: []
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 4
List is empty:
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 2
List is empty:
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 4
Invalid choice:
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 4

1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 1
Integer: 11
List after adding: [11]
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 1
Integer: 20
List after adding: [11, 20]
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 2
Integer: 26
Element not found:
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 2
Integer: 15
List after removing: [26]
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 2
Integer: 20
List after removing: []
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 4
List is empty:
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 2
List is empty:
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 4
Invalid choice:
1. Add:
2. Remove:
3. Display:
4. Quit:
Enter choice: 4

Test case 2
Passed

Average time
0.072 s
71.67 ms



Maximum time
0.089 s
89.00 ms



✓ 2 out of 2 shown test case(s) passed

✓ 1 out of 1 hidden test case(s) passed

Integer: 11

List after removing: [25]

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 2

Integer: 25

List after removing: []

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 3

List is empty

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 2

List is empty

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 8

Invalid choice

1. Add

2. Remove

Integer: 11

List after removing: [25]

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 2

Integer: 25

List after removing: []

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 3

List is empty

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 2

List is empty

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 8

Invalid choice

1. Add

2. Remove

< Prev

Reset

Submit

Next >

List is empty

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 8

Invalid choice

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 4

List is empty

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 8

Invalid choice

1. Add

2. Remove

3. Display

4. Quit

Enter choice: 4

< Prev

Reset

Submit

Next >